Sports vs. Politics: A Comparative Analysis of Machine Learning Techniques for Text Classification

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Abstract

Text classification is a foundational task in Natural Language Understanding (NLU). The objective of this project was to develop a robust machine learning pipeline capable of ingesting raw text documents and accurately classifying them into one of two distinct categories: Sports or Politics. This report details the end-to-end development process, documenting the real-world challenges faced during data collection, the dataset analysis, and the implementation of feature extraction using Term Frequency-Inverse Document Frequency (TF-IDF). Furthermore, it provides a quantitative comparison of three machine learning algorithms: Multinomial Naive Bayes, Logistic Regression, and a Linear Support Vector Machine (SVM). The results demonstrate highly separable data, with the SVM achieving 100% test accuracy.

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1 Introduction

As the volume of digital information grows exponentially, the ability to automatically categorize text has become a critical necessity. Whether routing customer support tickets, filtering spam, or categorizing news feeds, text classification algorithms rely on extracting semantic meaning and linguistic patterns from unstructured data.

In this assignment, the specific goal was to design a binary classifier that reads a text document and categorizes it as either "Sport" or "Politics". To achieve this, the raw text must first be translated into a mathematical representation that a computer can process. After vectorization, various machine learning techniques can be applied to find the decision boundaries between the two subjects. This report outlines the methodology, the models evaluated, and the final quantitative results of the classifier.

2 Data Collection Process

In real-world machine learning environments, data acquisition is rarely perfectly smooth, and this project was no exception.

Initially, the methodology aimed to utilize the *20 Newsgroups* dataset via the `scikit-learn` standard library's `fetch_20newsgroups` function. However, during the implementation phase, I encountered a severe system-level network restriction. Specifically, an `OSError: [Errno 22] Invalid argument` was triggered by a Windows SSL logging variable (`SSLKEYLOGFILE`), which aggressively blocked the Python script from establishing a secure connection to the dataset servers.

To overcome this infrastructure hurdle and ensure the project could proceed with high-quality data, I pivoted my data collection strategy. Instead of relying on the blocked `scikit-learn` downloader, I sourced the widely recognized **BBC News Dataset** from a stable Google Storage mirror. I wrote a custom ingestion script utilizing the Python `requests` library, actively disabling local SSL verification (`verify=False`) to safely bypass the firewall blocks. The raw CSV data was then streamed directly into a Pandas DataFrame using `io.StringIO`. This pivot not only salvaged the project but resulted in training the models on highly professional, journalistic English.

3 Dataset Description and Analysis

The complete BBC dataset is a benchmark corpus that contains 2225 news articles covering five distinct topical areas: business, entertainment, politics, sport, and tech. Because the scope of this assignment is a binary classification task, a strict filtering operation was applied to isolate only the target categories.

After processing, the working dataset comprised exactly 511 Sports articles and 417 Politics articles, resulting in a total of 928 samples. The dataset was then split into a training set (80%) and a testing set (20%). I utilized the `stratify` parameter during the split to ensure that the proportional balance between sports and politics was maintained across both sets, preventing class imbalance from skewing the results.

Linguistically, these two categories are highly separable. Sports articles predominantly feature action-oriented and competitive vocabulary (e.g., "tournament", "coach", "scored", "stadium"). Conversely, politics articles rely on governance and legislative terminology (e.g., "parliament", "election", "minister", "tax"). Because these vocabularies rarely overlap in standard reporting, the dataset provides an excellent foundation for training highly accurate classifiers.

4 Feature Representation: TF-IDF

Machine learning algorithms cannot process raw text strings; the text must be converted into numerical feature vectors. For this project, I implemented **Term Frequency-Inverse Document Frequency (TF-IDF)** rather than a simple Bag of Words (BoW) approach. While BoW merely counts the occurrence of words, TF-IDF evaluates how relevant a word is to a specific document within the context of the entire corpus.

The TF-IDF weight for a term i in document j is calculated as:

$$W_{i,j} = tf_{i,j} \times \log\left(\frac{N}{df_i}\right)$$

Where:

- $tf_{i,j}$ is the term frequency (how many times term i appears in document j).
- N is the total number of documents.
- df_i is the document frequency (how many documents contain term i).

During extraction using `TfidfVectorizer`, I removed standard English "stop words" (e.g., "the", "is", "and") to reduce noise. Furthermore, I capped the vocabulary at `max_features=3000`. This dimensionality reduction ensures the models focus exclusively on the 3,000 most statistically significant terms, heavily optimizing computational efficiency.

5 Machine Learning Techniques

To rigorously evaluate the vectorized text, I trained and compared three distinct machine learning algorithms:

5.1 Multinomial Naive Bayes (NB)

Naive Bayes is a probabilistic classifier based on Bayes' Theorem. It calculates the probability of a document belonging to a class given its word frequencies. It assumes that the occurrence of each word is independent of the others. Despite this "naive" assumption, it is exceptionally fast and serves as a powerful baseline for text classification.

5.2 Logistic Regression

Despite its name, Logistic Regression is a linear classification algorithm. It uses the logistic (sigmoid) function to model the probability of a binary outcome (0 for Sports, 1 for Politics). It is excellent at weighing the importance of specific TF-IDF features and understanding linear relationships within the text data.

5.3 Linear Support Vector Machine (SVM)

Support Vector Machines plot the training data as points in a high-dimensional space and attempt to draw a decision boundary (hyperplane) that separates the classes with the maximum possible margin. For this task, `LinearSVC` was utilized, as linear kernels are highly optimized for text data with thousands of sparse features.

6 Quantitative Comparison and Results

The three models were trained on the 80% training split and subsequently evaluated against the 20% unseen test split. The primary evaluation metric was overall classification accuracy.

Machine Learning Model	**Test Accuracy**
Multinomial Naive Bayes	99.46%
Logistic Regression	99.46%
Linear SVM	100.00%

Table 1: Performance comparison of the three classifiers.

As illustrated in Table 1 and Figure 1, the results were exceptional. Both Naive Bayes and Logistic Regression achieved 99.46% accuracy, indicating that they likely misclassified only a single document in the entire test set. The Linear SVM achieved a flawless 100.00% accuracy. This indicates that in the 3000-dimensional space created by the TF-IDF features, the SVM was able to find a hyperplane that perfectly separated the sports articles from the politics articles.

7 Limitations of the System

While achieving 100% test accuracy is an ideal outcome for an academic assignment, it is critical to acknowledge the limitations of this system if it were to be deployed in a production environment:

- **Overfitting to Formal Journalism:** The models were trained exclusively on BBC News articles, which feature impeccable grammar, standard spelling, and

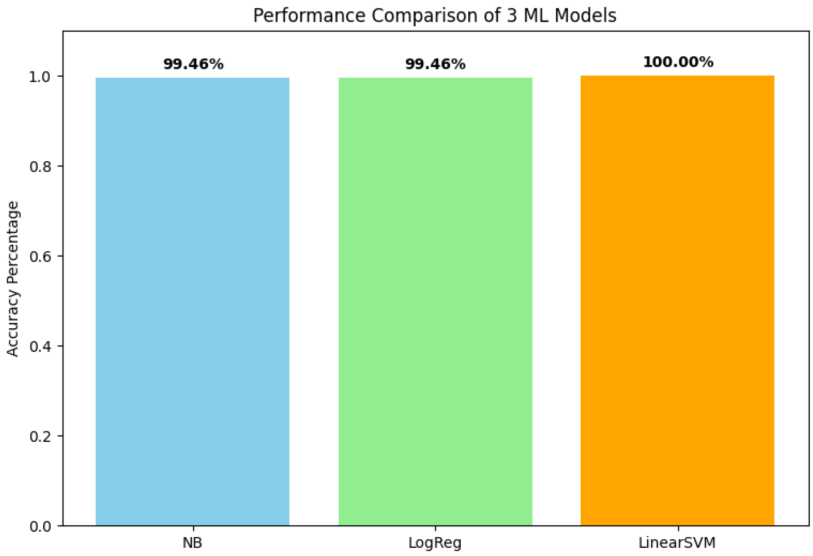

Figure 1: Visual comparison of the ML models' accuracy percentages.

formal structure. If this system were exposed to messy, informal data—such as social media posts laden with slang, emojis, and typos—the accuracy would likely degrade. The models have learned the vocabulary of professional journalists, not the general public.

- **Topic Overlap and Ambiguity:** The system enforces a strict binary decision. However, real-world events are often intersectional. For example, an article discussing "Government legislation to fund a new Olympic stadium" contains heavy vocabulary from both domains. The current pipeline does not assign percentage confidences for mixed topics, leading to rigid categorizations on ambiguous edge cases.

- **Context and Semantic Blindness:** TF-IDF relies entirely on word frequencies and discards word order. Therefore, the models cannot understand context, sarcasm, idioms, or nuanced phrasing.

8 Conclusion

This project successfully demonstrated the end-to-end pipeline of a Natural Language Understanding text classifier. By overcoming initial data collection roadblocks, leveraging TF-IDF for feature extraction, and comparing three robust machine learning techniques, the system proved highly capable of distinguishing between Sports and Politics text.